

Umer Niazi

Lahore, Pakistan | +92 339 213 3364 | umer.niazi@proton.me | linkedin.com/in/umer-niazi | github.com/umer-niazi | umer-niazi.github.io

EDUCATION

University of Engineering and Technology, Lahore

Lahore, Pakistan

BS in Artificial Intelligence (CGPA: 3.30/4.00)

2025 – 2029 (Expected)

Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, Software Engineering, Computer Networks, Programming for AI, Applied Statistics and Probability, Linear Algebra, Discrete Mathematics

Forman Christian College

Lahore, Pakistan

Intermediate in Computer Science (ICS)

2022 – 2024

PROJECTS

News NLP Pipeline [\[code\]](#) [\[demo\]](#) | *Python, BERTopic, spaCy, Transformers, Streamlit, SQLite*

May 2026

- Built an end-to-end NLP pipeline over 15 years of Dawn News headlines for topic discovery, sentiment analysis, and named-entity extraction, with no manual labeling
- Scraped and processed 350K+ headlines, extracted 420K+ entity mentions with spaCy, and modeled 20 distinct topics with BERTopic; deployed an interactive Streamlit dashboard
- Found cricket (~32.5K headlines) and legal cases (~29.2K) to be the largest topics, average sentiment slightly negative (−0.085), and the US the second most mentioned entity after Pakistan, ahead of India

Himalayan Wildlife Classifier [\[code\]](#) [\[demo\]](#) | *Python, PyTorch, ResNet18, Streamlit*

July 2026

- Built an image classifier for Snow Leopard, Markhor, and Himalayan Brown Bear (819 images across the three species) using transfer learning on a frozen ResNet18 backbone
- Extended the first version with a background Other class and an 80% confidence threshold for selective prediction to filter ambiguous inputs; achieved 96.55% validation accuracy and deployed an interactive Streamlit demo

wmus [\[code\]](#) | *Python, windows-curses, pygame*

June 2026

- Built a lightweight terminal-based music player for Windows inspired by the cmus workflow, featuring keyboard-first navigation, a custom curses interface, and configurable keybindings
- Implemented fuzzy track search and metadata caching to handle large audio collections; packaged as a standalone installer using PyInstaller and Inno Setup

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL, GDScript

AI & Machine Learning: PyTorch, scikit-learn, Transformers, spaCy, BERTopic, Pandas, NumPy, Computer Vision, NLP

Software & Systems: Streamlit, SQLite, Linux, Git, Bash

Game Dev: Godot, Unity